Sai Srinivas Pedhapolla

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Professional Summary

Entry-level Data Engineer and Data Science graduate with internship experience designing and operating Python-based data pipelines, ETL/ELT workflows, and backend data services. Demonstrated ability to ingest, model, and validate structured and semi-structured data from APIs, files, and relational databases for analytics and machine learning use cases. Hands-on experience automating Linux-based data workflows using Python and Bash, with strong focus on logging, monitoring, and data quality. Proficient in SQL, data modeling, schema management, and version control using Git across collaborative, cloud-oriented environments. Eager to contribute to large-scale data engineering projects by building reliable, scalable pipelines and partnering with cross-functional teams to unlock data-driven insights.

Skills

- **Programming & Querying**: Python, SQL, Bash, C++, JavaScript, Java, C#, Pandas, NumPy, HTML/CSS (basic frontend), JavaScript for web frontends
- **Data Engineering & Pipelines**: ETL/ELT workflows, data ingestion, incremental loads, data validation, schema management, reusable pipeline modules, Data Modeling, data warehousing concepts, end-to-end ETL/ELT pipeline design
- **Machine Learning & AI**: Machine Learning fundamentals, TensorFlow, data preprocessing, feature engineering, model experimentation, exploratory data analysis (EDA), statistical data analysis
- **Software Engineering**: Object-Oriented Design, modular software development, debugging & troubleshooting, performance optimization
- **Systems & Platforms**: Linux / Unix environments, macOS debugging, cloud-based platforms, OS-level scripting
- **Cloud & Data Platforms**: AWS (S3, Glue – academic/project exposure), GCP, Snowflake, Firebase
- **Databases**: SQL Server (academic exposure), PostgreSQL, Snowflake, MongoDB
- **Data Lake & Analytics**: Parquet (project exposure), partitioning strategies, large-scale dataset processing, Big Data Technologies, big data processing, distributed data processing concepts
- **Tools & DevOps**: Git/GitHub, Linux, CI/CD fundamentals, Power BI, Tableau, Data Visualization Tools, Jupyter Notebook, VS Code / IDE-based development

Work Experience

Goods Unite Us **Sep 2025 - Present**
Software Intern *NJ, USA*

- Engineered Python-based data pipelines and backend services with robust data modeling to support advanced analytics and machine learning workloads, integrating relational databases and cloud storage to deliver reliable batch and near real-time data feeds to downstream consumers.
- Architected modular ETL/ELT components to ingest, transform, and model structured and semi-structured data using object-oriented design and big data technologies, optimizing pipeline performance through partitioning, incremental loads, and schema evolution strategies.
- Automated high-volume data ingestion and validation by scripting Linux-based workflows in Python and Bash, implementing robust logging, alerting, and error handling to improve data quality, throughput, and recovery time for failed jobs.
- Prepared, cleaned, and transformed large datasets for machine learning models by standardizing preprocessing, feature engineering, and data quality checks, enabling reproducible experiments and scalable model training pipelines.
- Collaborated with cross-functional data, ML, and engineering teams to diagnose production data issues, performing root-cause analysis and implementing permanent fixes that increased data pipeline reliability and reduced incident frequency.
- Leveraged Git-based workflows for version control, code reviews, and CI/CD integration across multiple data engineering codebases, contributing to automated testing, linting, and deployment pipelines for data services and APIs.

Webdaddy **Aug 2024 - Feb 2025**
Python Developer R&D Specialist Intern *India*

- Developed Python-based data processing tools and utilities to build end-to-end data pipelines, integrating APIs, flat files, and relational databases to create unified analytical datasets for reporting and machine learning use cases.
- Diagnosed and resolved performance, reliability, and data integrity issues across data workflows by applying systematic debugging, query tuning, and log analysis, reducing pipeline runtimes and eliminating recurring data discrepancies.
- Implemented reusable, object-oriented Python modules for ETL/ELT workflows and transformation logic, establishing shared libraries, documentation, and coding standards that improved maintainability and developer productivity.
- Automated end-to-end system and data workflows using Python and Bash, orchestrating scheduled jobs with cron and task schedulers to increase data pipeline throughput and minimize manual operational effort.
- Collaborated with research and engineering teams on machine learning experiments, feature engineering, and training data pipelines, designing experiment-ready datasets and tracking key metrics to support rapid model iteration.

Findem, Inc. **Jul 2023 - Dec 2023**
R&D Data Analyst Intern *Bengaluru, India*

- Engineered backend services using FastAPI, Django, and Java microservices to support scalable data ingestion, transformation, and API-based data access, exposing secure endpoints that powered internal analytics and partner integrations.
- Maintained and optimized Linux-based data processing applications, tuning resource usage, parallelism, and observability dashboards to enhance data throughput, reliability, and system uptime.
- Prepared and transformed datasets for machine learning by implementing feature engineering, data cleansing, and validation checks, helping ensure high-quality training data and consistent experimental pipelines.
- Utilized Git-based workflows for collaborative development, code reviews, and version control across data engineering and analytics projects, following branching strategies that reduced integration conflicts and improved code quality.

- Supported cloud-hosted analytics and ML systems using C#, integrating with cloud data storage, monitoring, and reporting components, and contributing to optimization and migration efforts on modern data platforms.

Projects

Scalable Data Ingestion & ETL Pipeline

- Designed and implemented Python-based systems software to ingest and process structured datasets.
- Applied modular, object-oriented design for scalability and maintainability.
- Implemented validation, logging, and debugging mechanisms in Linux environments.
- Supported downstream machine learning model experimentation.

Automated Analytics & Reporting Pipelines

- Built automation tools using Python and Bash to streamline data preparation for ML pipelines.
- Integrated software components with visualization and reporting layers using JavaScript-based dashboards.

Education

New Jersey Institute of Technology **2024 - 2025**
Master of Science, Data Science *Newark*

Sri Chandrasekharendra Saraswathi Viswa Mahavidyalaya **2019 - 2023**
Bachelor of Science, Computer Science